Catherine Hae Seung Jeon

Google Scholar — GitHub — LinkedIn — Website

RESEARCH INTERESTS

❖ **Efficient Deep Learning**
- *Few-shot Learning, Continual Learning*

❖ **Representation Learning**
- *Metric Learning, Embedding Model*

❖ **Robust Deep Learning**
- *Noise/Out-of-distribution/Adversarial Robustness*

❖ **AI for Social Good**
- *AI for Science/Healthcare/Security*

EDUCATION

Ewha Womans University — *Seoul, South Korea*
❖ ***M.S. in Computer Science & Engineering (Advisor: Prof. S.E. Oh)*** — *09/2024 — 08/2026 (Expected)*
- GPA: overall 4.26/4.30

Ewha Womans University — *Seoul, South Korea*
❖ ***B.S. in Computer Science & Engineering*** — *03/2020 — 08/2024*
- GPA: overall 4.15/4.30
- Honors: Dean's List (7 out of 8 semesters), Summa Cum Laude

PUBLICATIONS

Conference Publications

Jeon, Haeseung*, Kim, S.*, Mathews, N., Kang, H., & Oh, S.E. (2026). "RoFiRe: Robust Website Fingerprinting on Real-World Tor Traffic via Improved Augmentation and Normalization." To appear in *The 2026 ACM Web Conference (**WWW**)*. [CORE A*, Acceptance Rate = 20%]

Hwang, C.*, **Jeon, Haeseung***, Hong, J., Kang, H., Mathews, N., Kim, G. & Oh, S.E. (2025). "Enhancing Search Privacy on Tor: Advanced Deep Keyword Fingerprinting Attacks and BurstGuard Defense." In *Proceedings of the 20th ACM Asia Conference on Computer and Communications Security (**ASIACCS**)*. [CORE A, Acceptance Rate = 20%]

Hwang, C.*, **Jeon, Haeseung***, Kim, G., Hong, J., Kang, H., & Oh, S.E. (2024). "DKF: Employing Deep Learning for Keyword Fingerprinting Attacks on Tor." In *Proceedings of the Korea Computer Congress (**KCC**)*. [Short Paper] [Distinguished Paper Award]

Workshop/Poster Publications

Jeon, Haeseung, Hong, J., Hong, S., Kang, H., Kim, B., Oh, S. E., & Kim, N. (2025). "Domain-Adapted Automatic Speech Recognition with Deep Neural Networks for Enhanced Speech Intelligibility Prediction." In *Proceedings of the 6th Clarity Workshop on Improving Speech-in-Noise for Hearing Devices (**Interspeech Workshop**)*. [Short Paper] [Challenge Top 5]

Hwang, C*, **Jeon, Haeseung***, Kim, G., Hong, J., Kang, H., & Oh, S.E. (2024). "Securing Search Privacy on Tor: Deep Keyword Fingerprinting and BurstGuard Defense." Presented at *The 40th Annual Computer Security Applications Conference (**ACSAC**)*. [Poster]

Under-Review Journal/Conference Publications

Asadi, Z., **Jeon, Haeseung**, Han, S., Kamol, M. M., Oh, S.E. & Rahman, M. S. (2026). "FreeMOCA: Memory-Free Continual Learning for Malicious Code Analysis with Mode Connectivity and Interpolation." Under review at *The 43rd International Conference on Machine Learning (**ICML**)*.

Kim, N., **Jeon, Haeseung**, & Bai, M. R. (2026). "Speakers as Intelligent Sensors: Impedance-Based Acoustic Load Inference Powered by Deep Learning." Under review at *The IEEE Sensors Journal (**SENSORS**)*.

Park, J., **Jeon, Haeseung**, Ji, A., Piplai, A., Rahman, M. S., & Oh, S.E. (2025). "TabCL: Continual Malware Classification with Tabular-Aware Generation." Under review at *The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (**PAKDD**)*.

*Equally contributing authors, listed in alphabetical order.

RESEARCH EXPERIENCE

Purdue University | Sensor, Electro-Acoustics Technology Lab *Indiana, United States*

❖ ***Visiting Researcher (Research Advisor: Prof. N. Kim)*** *01/2026 — 05/2026 (Expected)*

- Building a physical basis (acoustic load, temperature, and humidity) prediction model with impedance sensing from electroacoustic edge devices. Achieved near-perfect accuracy, demonstrating the potential of DL as an effective and efficient diagnostic tool.

❖ ***Part-Time Researcher*** *05/2025 — 12/2025*

- Directed Transformer-based speech clarity prediction model development to support hearing aid users. Suggested fine-tuning an ASR backbone with speech-in-noise audio and masked transcription for inaudibility simulation, and achieved 5th place in the challenge.

Ewha Womans University | AI Security Lab *Seoul, South Korea*

❖ ***Graduate Research Assistant & Lab Head (Advisor: Prof. S.E. Oh)*** *09/2024 — 06/2026 (Expected)*

- Proposed a robust few-shot model using LLM-inspired normalization (pre-norm, RMSNorm) and dynamic window-based augmentation, achieving up to 9% gains over SimCLR-based SOTA across few-shot, concept drift, and open-set settings without large-scale pre-training.
- Leading advanced time-series embedding model development utilizing metric learning models and FENs in collaboration with the University of Edinburgh and the U.S. NRL.

❖ ***Undergraduate Research Assistant*** *01/2023 — 08/2024*

- Led research on the first DL-based network traffic fingerprinting model, achieving a 41-55% gains over a previous SOTA. Built a Selenium crawler to collect 300K+ fresh samples over 1+ years, profiled unique data patterns, and benchmarked ML/DL baselines on a new dataset.

UTEP | Intelligent & Quantum Secure Advanced Cyber Defense Lab *Texas, United States*

❖ ***Visiting Researcher (Research Advisor: Prof. M.S. Rahman)*** *06/2025 — 08/2025*

- Built a continual learning framework with a conditional tabular GAN and improved storage/training efficiency by generative replay. Drove experiments and achieved up to 16% improvements over the baselines (GAN and TVAE) in Class-IL/Time-IL scenarios.

PROJECTS

Seoul National University | THUNDER Research Group *GitHub*

❖ ***Accelerator Programming Bootcamp Participant*** *02/2025*

- Led re-implementation of the CPU-based MoE model with custom CUDA kernels to enable GPU-based parallelism. Analyzed bottlenecks using Nsight Systems, applied advanced optimization techniques such as CUDA streams, kernel fusion, and warp occupancy optimization. Achieved a 650× throughput improvement and a 3rd place award in the competition.

INDUSTRY EXPERIENCE

Samyang Data Systems Inc. | Cloud Solutions Team *Seoul, South Korea*

❖ ***Cloud Engineer Intern*** *03/2023 — 06/2023*

- Implemented a Golang monitoring agent and E2E pipeline that collects, stores, and visualizes real-time metrics from AWS infrastructure by utilizing InfluxDB and Grafana. Deployed agents to 10+ client servers, ensuring software reliability through stress tests.

AWARDS & SCHOLARSHIPS

EWU Research Assistant Scholarship — *half-tuition*	*FA 2024*
EWU Outstanding Ewha Scientist Admissions Scholarship — *half-tuition (2 years)*	*FA 2024 — SP 2026*
Korea Computer Congress Distinguished Paper Award (top 6%)	*SP 2024*
EWU National Program of Excellence in Software Scholarship — *KRW ₩1M*	*SP 2024*
Huawei ICT Talent Development Scholarship — *KRW ₩1M*	*FA 2023*
EWU Excellence Honors Scholarship (top 6%) — *quarter-tuition*	*FA 2023*
EWU Highest Honors Scholarship (top 2%) — *half-tuition*	*FA 2020, SP 2022, SP 2023*
EWU W4 Admissions Scholarship (top entrant) — *full-tuition (4 years)*	*SP 2020 — SP 2024*

COMMUNITY ENGAGEMENT

EWU CS Undergraduate Lab Pair, Invited Speaker	*09/2025*
UTEP CS Summer Research Symposium, Invited Speaker	*08/2025*
EWU CS Undergraduate Open Lab Seminar, Invited Speaker	*03/2025*
GDSC Devfest π 2024: Responsible AI in action, Invited Speaker	*03/2024*
GDSC Devfest Cloud 2023, Organizer	*12/2023*
AWS Cloud Club (ACC), Cloud Team Executive	*09/2023 — 08/2024*
EWU Undergraduate Peer Tutoring Program, Lecturer	*06/2022, 06/2024*
Google Developer Student Clubs (GDSC), Backend Team Executive	*09/2022 — 07/2024*